Processing and Analysis of Lidar Data for Bat Detection Using Classical Methods and Artificial Intelligence

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Offen im Denken

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University of Duisburg-Essen

Motivation

Problems

- Bat mortality at wind turbine installations
- Collision and barotrauma effects
- Fatality rates exceed ecological sustainability

Challenges

- Increasing conflict between wind energy and wildlife
- Conservative curtailment
- Reduced turbine availability
- Lack of reliable real-time information

Approaches

- Automated detection of airborne wildlife
- Real-time monitoring
- Targeted and adaptive curtailment

[https://chatgpt.com/]

Contents

- Motivation
- Background and Related Work
- Methods
- System Overview and Processing Pipeline
- Results
- Discussion and Future Work

Background and Related Work

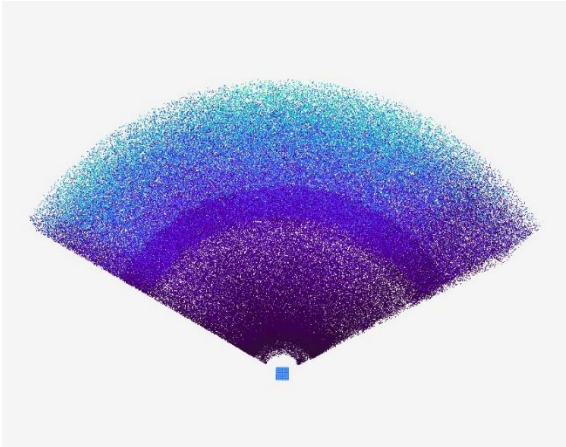
- Acoustic monitoring [Krapivnitckaia, 2024; Arlt, 2021]
  - Species classification via echolocation
  - Limited detection range
  - No reliable spatial localization
- Thermal and radar-based systems [Krapivnitckaia, 2024]
  - Improved detection capability
  - Require complementary validation
  - High hardware and operational cost
- LiDAR-based approaches for bat activity monitoring [Werber, 2023]
  - Demonstrated 3D tracking potential
  - Combined with acoustic or radar systems
  - No standalone real-time turbine automation
- Combined LiDAR–NIR sensing remains underexplored for turbine-level monitoring

➔ Research gap in automated, spatially accurate, real-time bat detection

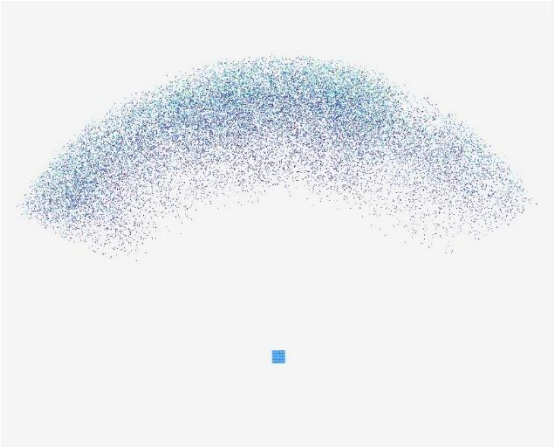
Preprocessing

Noise Filtration

- Raw LiDAR point clouds
  with isolated and spurious returns
- Need for noise suppression
  to preserve small airborne objects
- Evaluation of noise filtering strategies
  - Intensity-based filtering
  - Radius Outlier Removal (ROR)
  - Dynamic Radius Outlier Removal (DROR)
- Range-dependent point sparsity in outdoor scenes



Point cloud before intensity filter

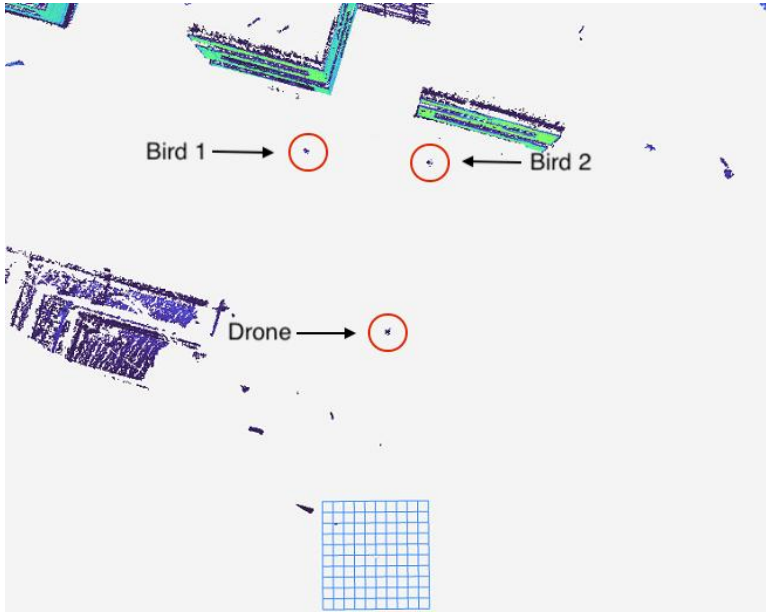


Point cloud after intensity filter

Preprocessig

Static Background Removal

- Dominant static structures in outdoor LiDAR scenes
- Background subtraction for isolation of temporally varying foreground points
- Implemented voxel-based background subtraction approaches
  - 3D density threshold filtering (Radial zone modelling)
  - Voxel confidence learning filtering (Per-voxel confidence score)
  - Occupancy ratio filtering (Recursive octree)
- Methods differ in spatial adaptivity and computational cost

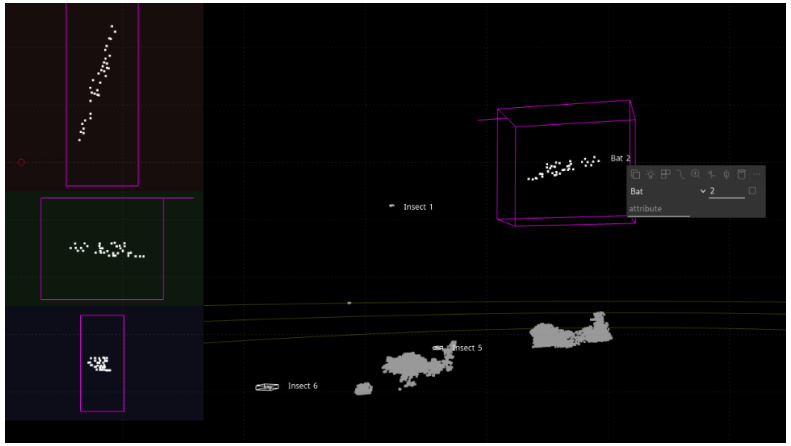


Raw

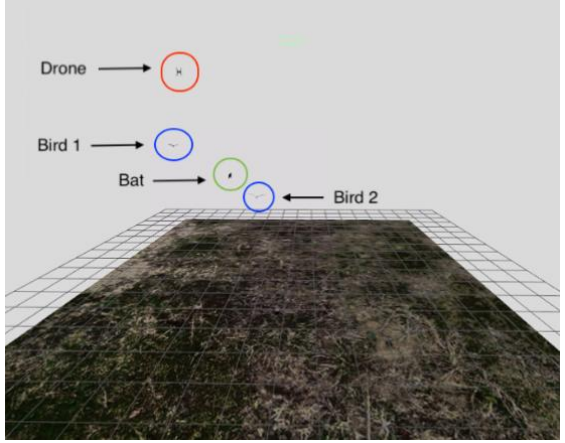
Object Detection

Overview

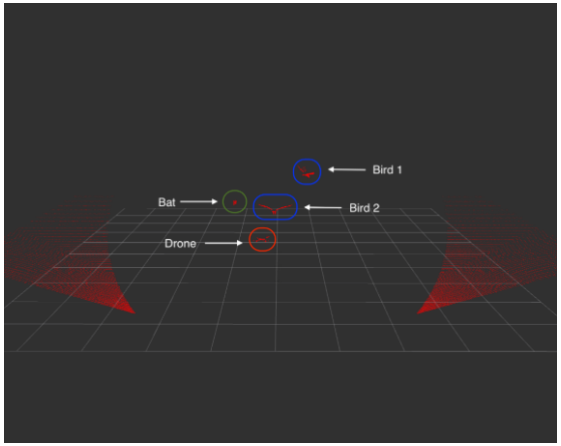
- Grouping of preprocessed LiDAR foreground points into candidate objects
- Unsupervised object detection via clustering in three-dimensional space
- Evaluated clustering algorithms on simulated and real-world datasets
  - DBSCAN
  - HDBSCAN
  - K-Means
  - Mean Shift



Real Dataset Frame



Gazebo Simulation Frame

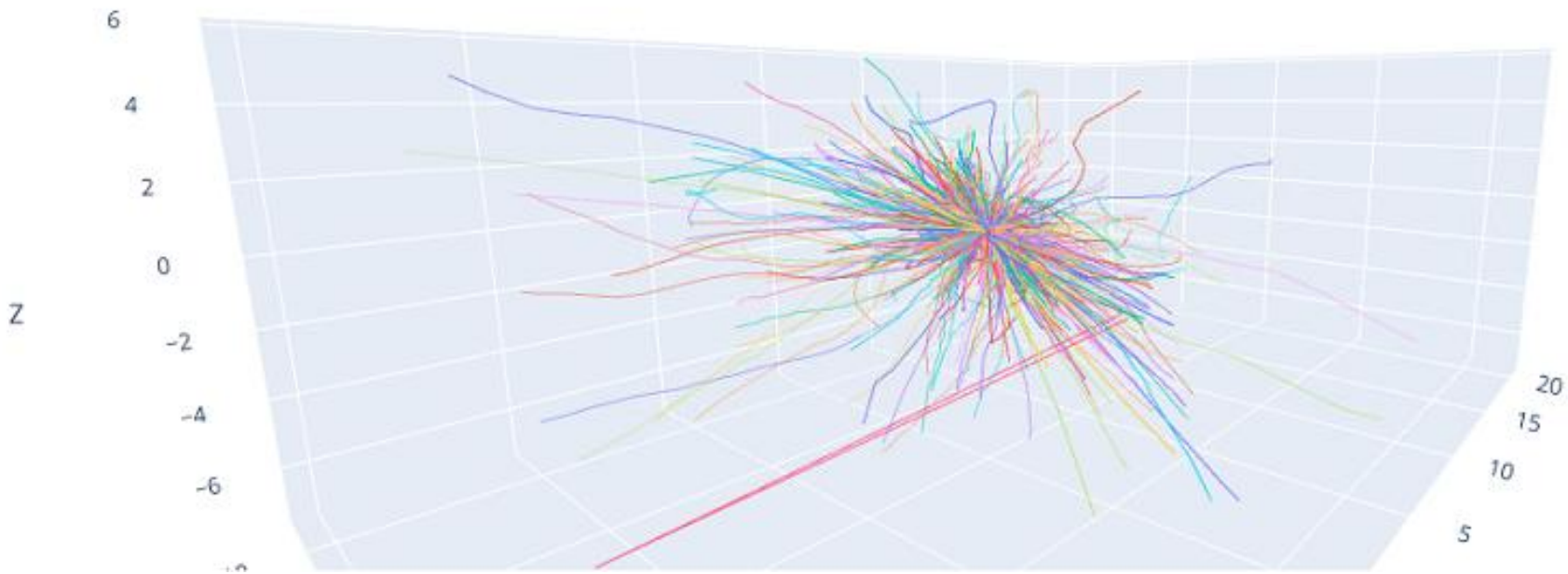


Simulation Point Cloud

Object Classification

Overview

- Trajectory-level classification instead of frame-level analysis
- Focus on spatio-temporal motion patterns over species-specific appearance cues
- No semantic or species labels for real-world data
- Motion pattern separability in latent space

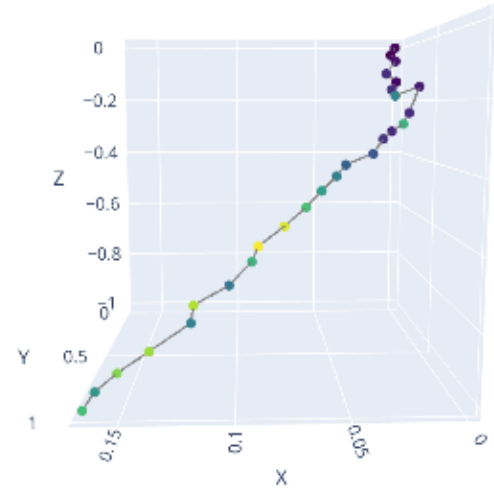


Extracted object trajectories

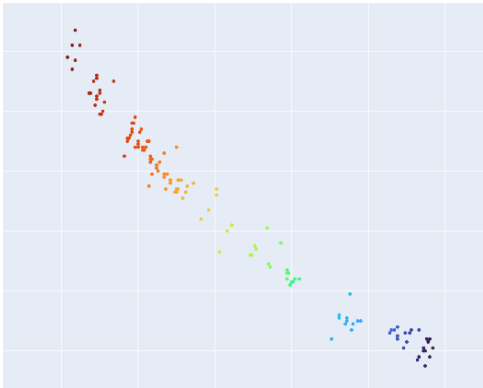
Object Classification

Trajectory Generation

- Linking of object detections across frames via tracking
- Kalman filtering for consistent object identities over time
- Manual correction of identity switches and broken tracks
- Extracted trajectory representations from corrected tracks
  - Centroid-based trajectory representation
  - Spatio-temporal Bird's eye view (BEV) trajectory representation



Centroid-Based Trajectory Example

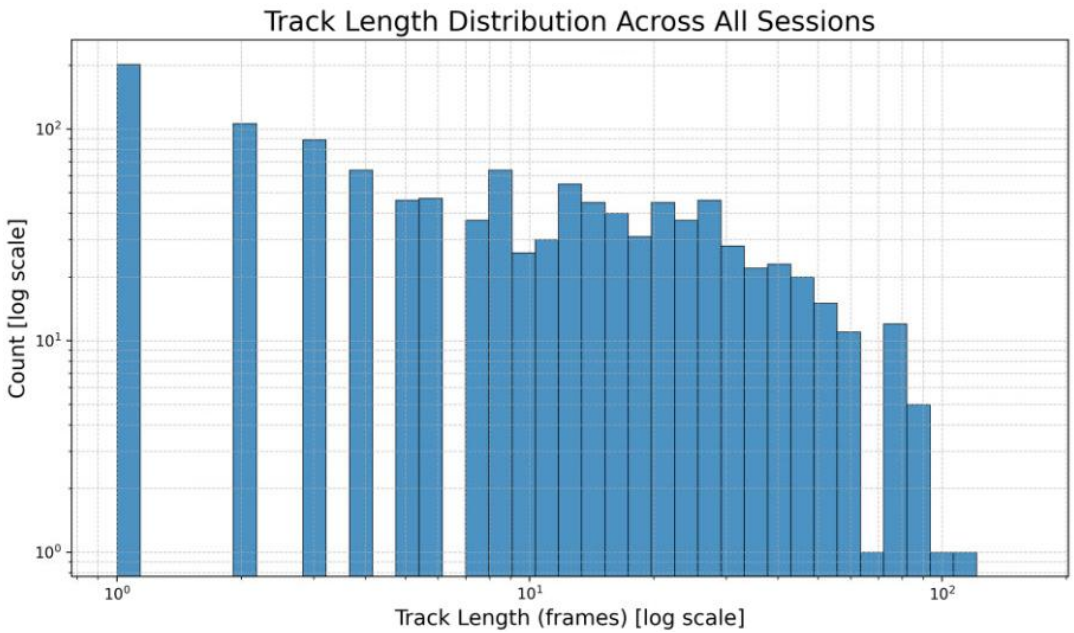


Spatio-Temporal BEV Trajectory Example

Object Classification

Motivation for Synthetic Data

- Short and sparse field-recorded trajectories
- Total of 378 unique object trajectories
  - Trajectory length statistics
  - Median length of 2 frames
  - Mean length of approximately 8 frames
- Limited time observation for motion analysis
- Absence of ground-truth labels for real-world trajectories
- Controlled and interpretable motion patterns from synthetic data

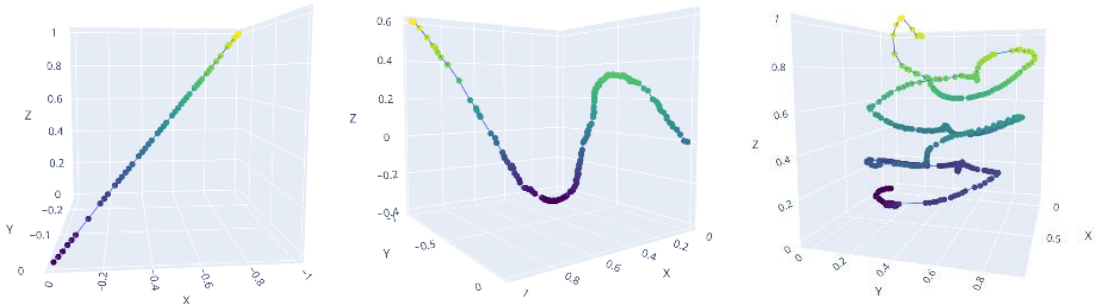


Distribution of object track lengths across all field-recorded sessions

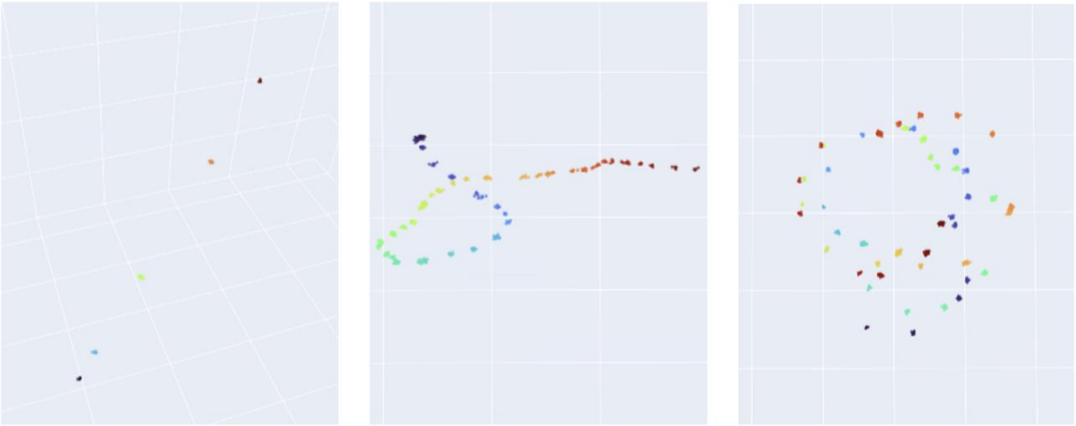
Object Classification

Synthetic Dataset

- Synthetic trajectories from drone-based data collection
- Motion paths from predefined geometric flight plans
- Recorded trajectory types
  - Straight-line motion
  - Circular motion
  - Random motion
- Extended trajectories for motion-pattern analysis



Centroid-Based Synthetic Trajectory

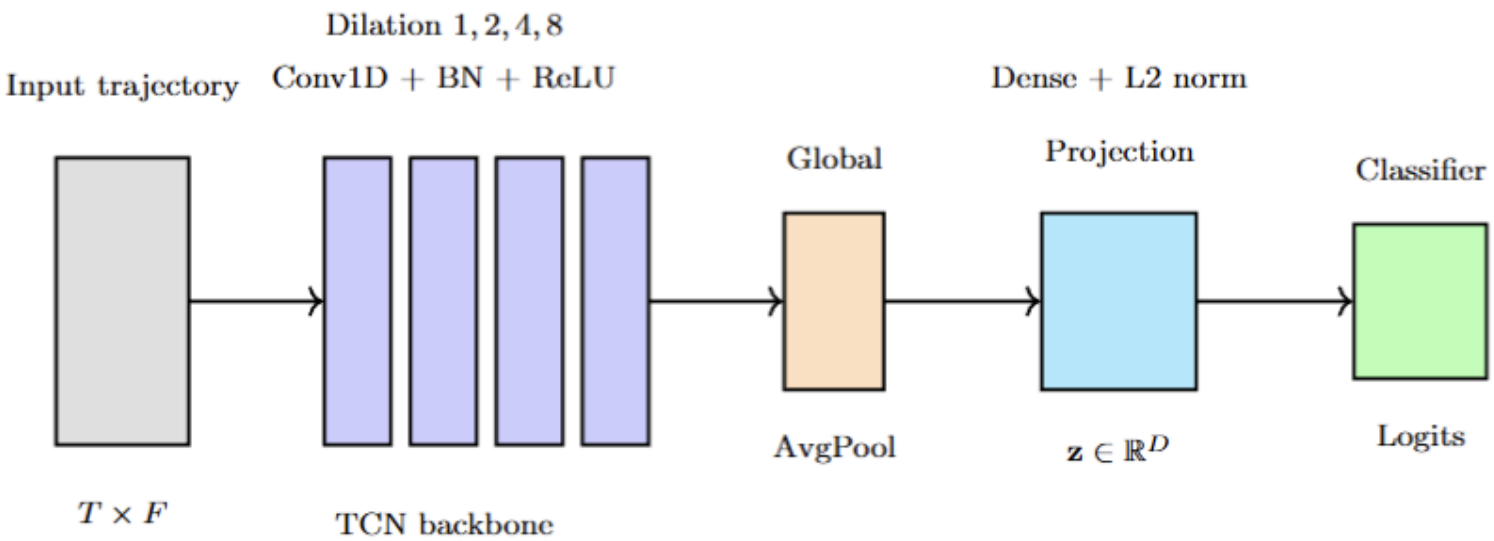


Spatio-Temporal Synthetic Trajectory

Object Classification

Supervised Reference Experiment

- Supervised learning as reference on synthetic trajectories
- Motion-pattern labels available for straight circular and random trajectories
- Temporal convolutional network trained on centroid-based trajectories

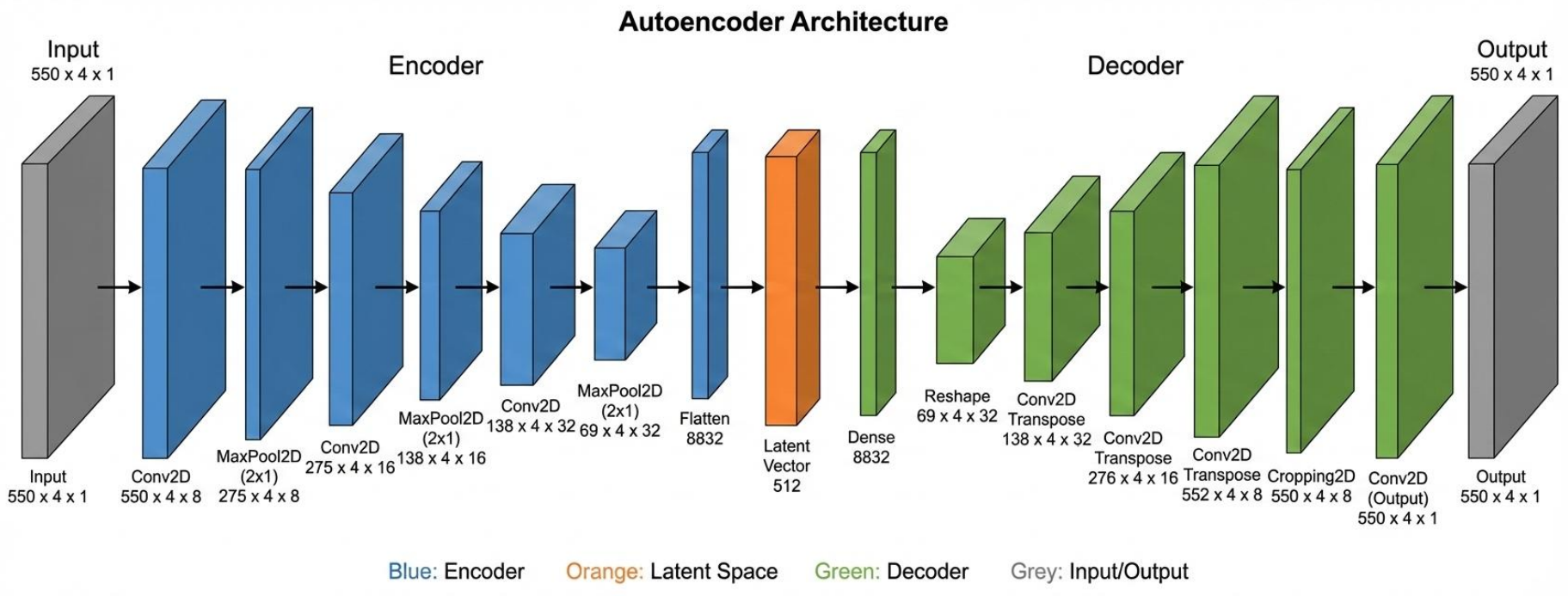


TCN Architecture

Object Classification

Unsupervised Centroid-Based Representation

- Unsupervised convolutional autoencoder trained on centroid trajectories
- Input encodes object motion using position and intensity over time
- Model learns compact latent representations without labels

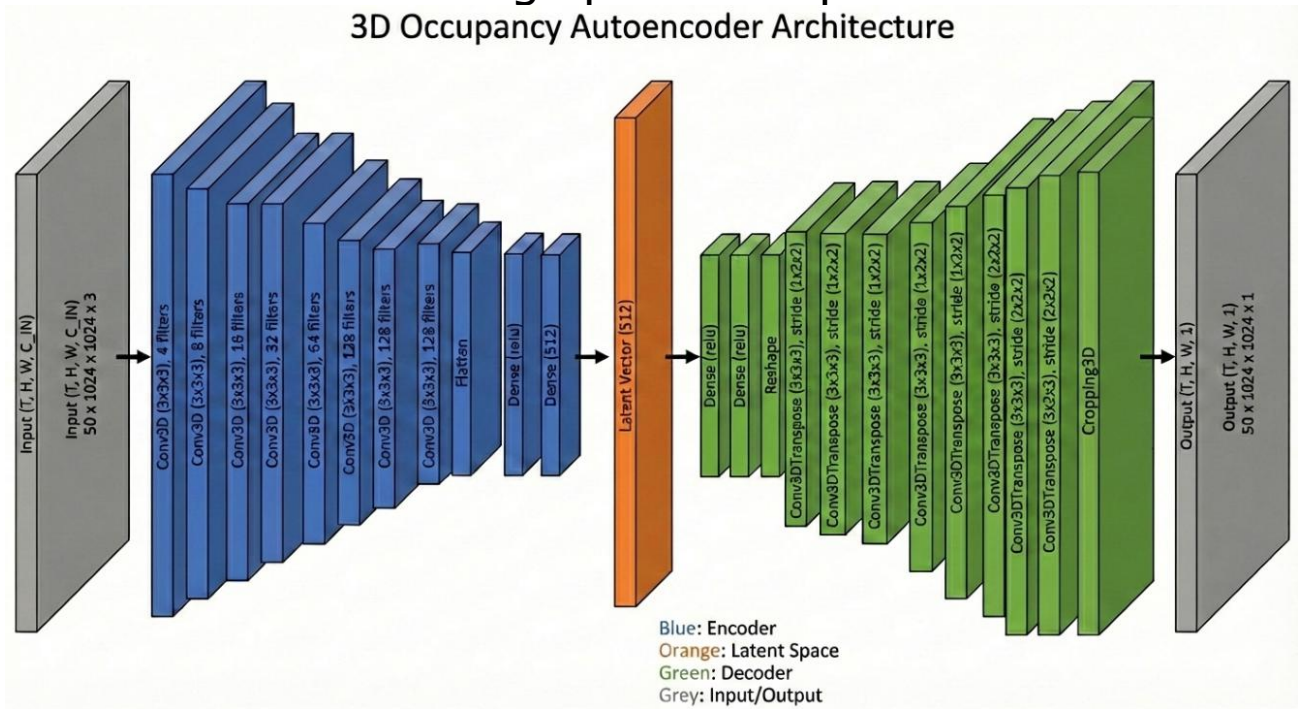


Autoencoder Architecture

Object Classification

Unsupervised Spatio-Temporal BEV Representation (Model A)

- Full object point cloud preserved and projected to BEV
- Trajectories encoded as spatio-temporal BEV voxel sequences
- Three-dimensional convolutional autoencoder used
- Model trained using spatio-temporal reconstruction objective

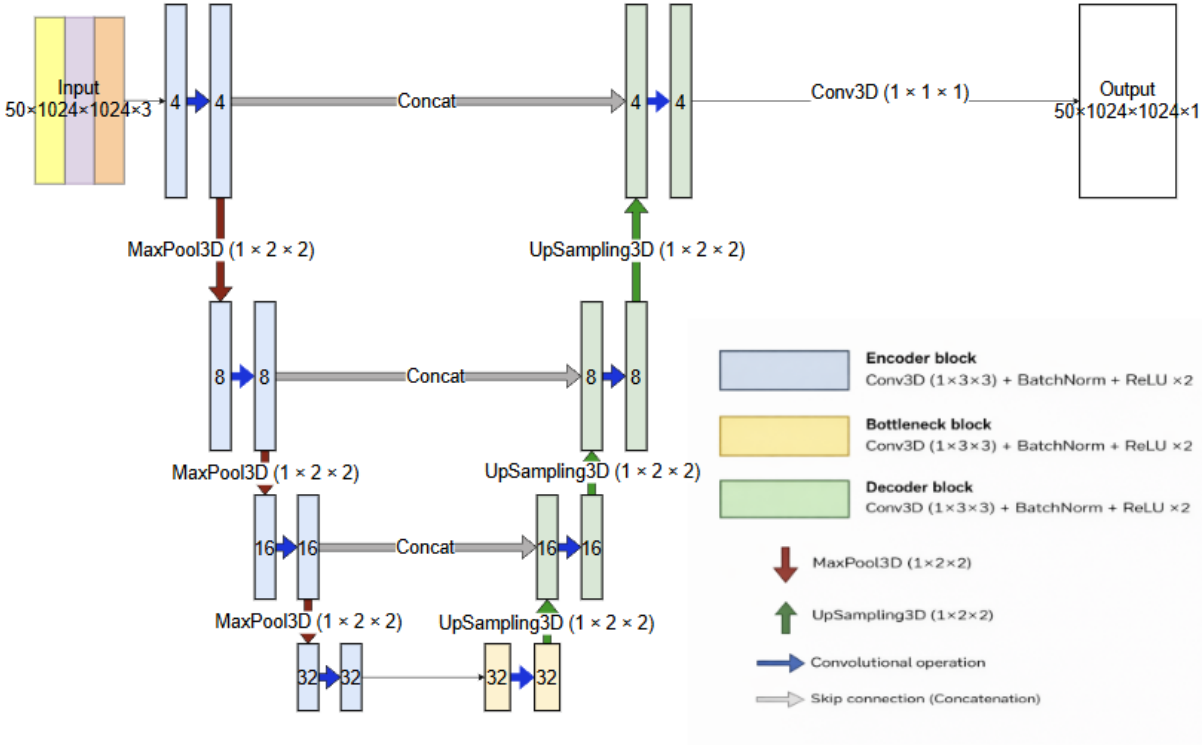


Autoencoder Architecture

Object Classification

Unsupervised Spatio-Temporal BEV Representation (Model B)

- 3D U-Net for spatio-temporal occupancy prediction
- Loss design handles extreme class imbalance
- Occupancy-focused learning with feature representation analysis

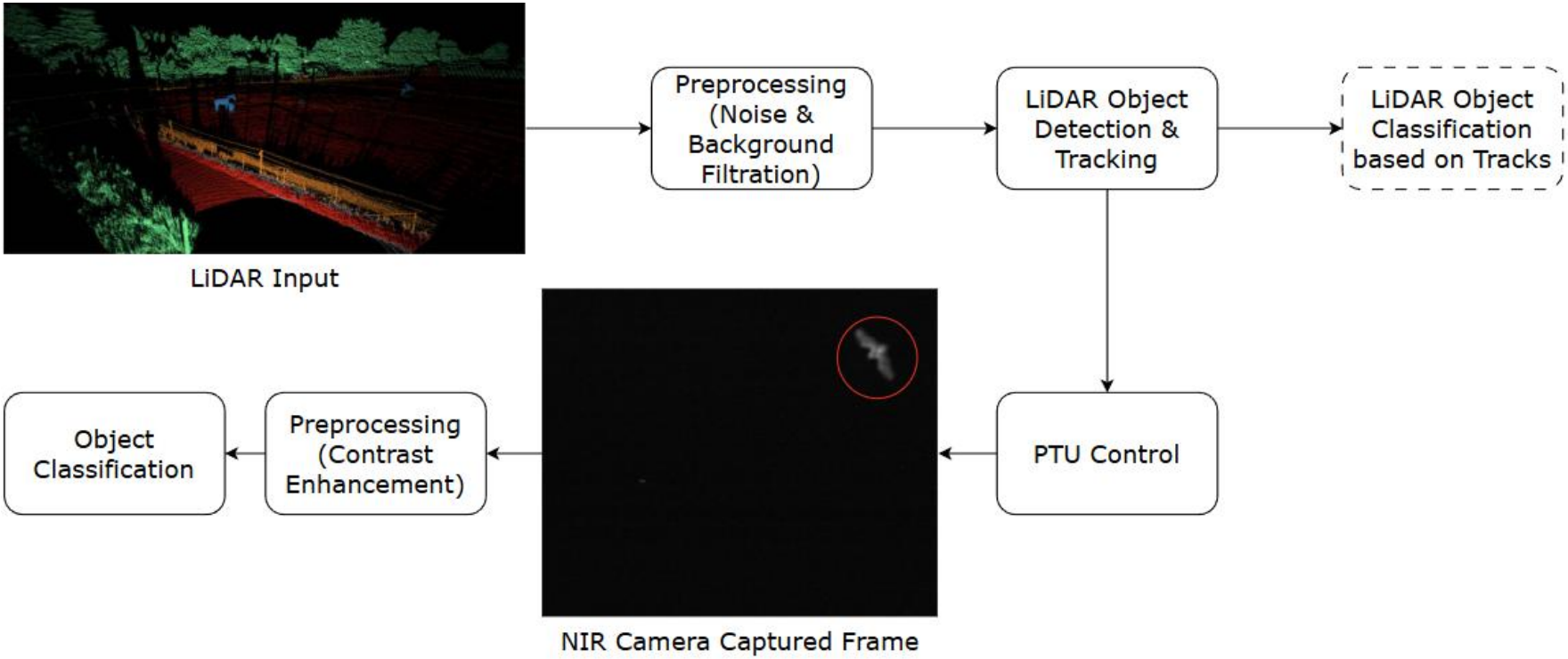


U-NET Architecture

System Overview

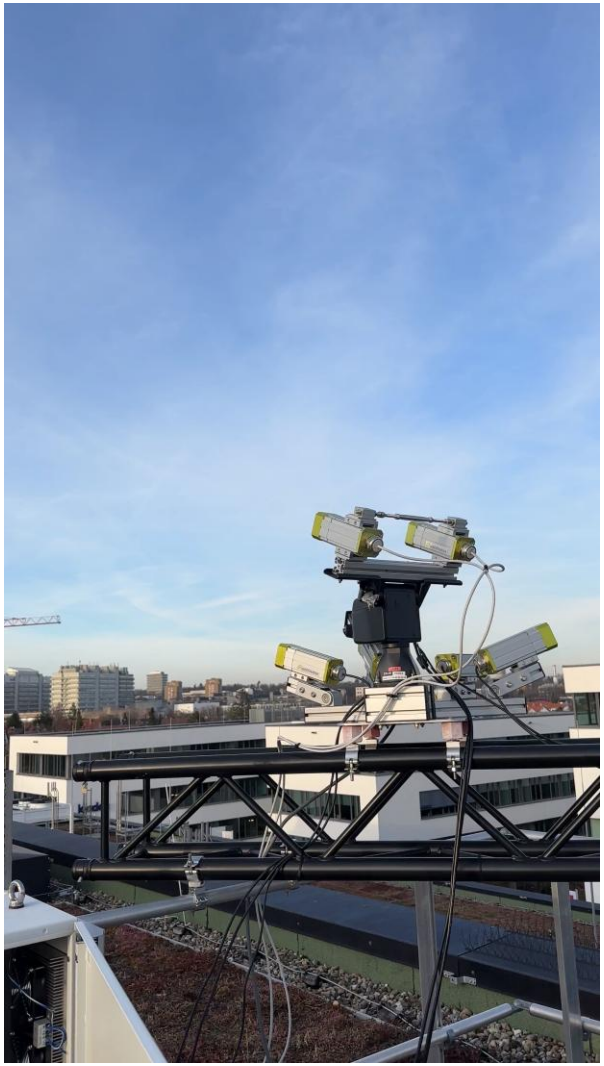
LiDAR-Based Detection with PTU-Controlled NIR Support

- LiDAR-based object detection and tracking
- PTU-controlled NIR alignment for active observation
- Secondary NIR cameras for synchronized recording



Overview of the proposed LiDAR-NIR system architecture and processing pipeline

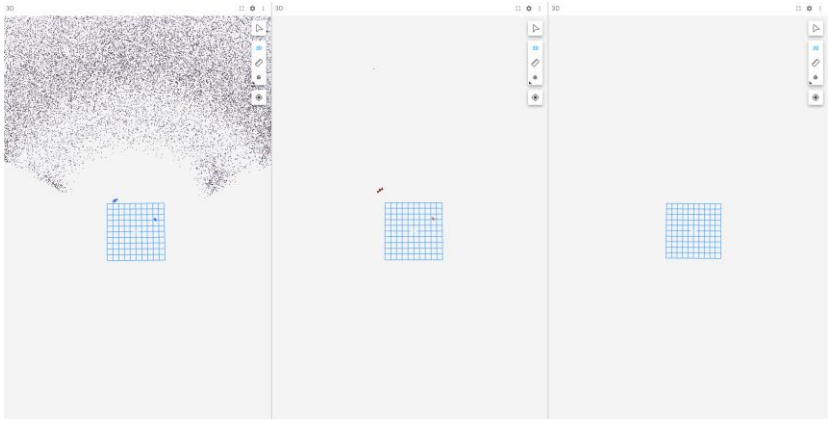
System Overview



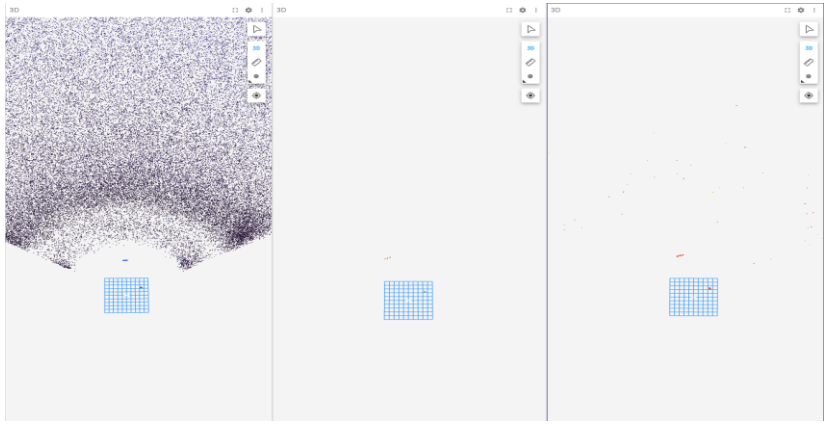
Results

Noise Filtration

- ROR may remove distant targets due to sparse point distribution
- DROR adapts to range-dependent density variations
- Intensity filtering depends on reflectance thresholds
- Small objects risk removal when intensity overlaps with noise
- DROR preserves cluster continuity and improves throughput



Raw ROR Intensity-Based



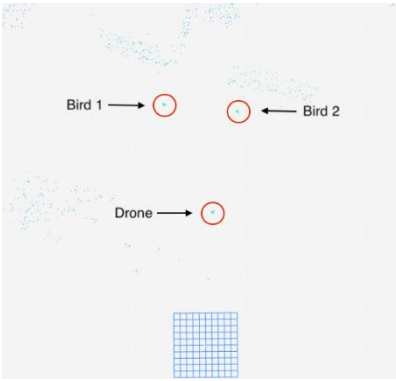
Raw ROR DROR

➔ Robust noise suppression with preservation of sparsely sampled airborne objects

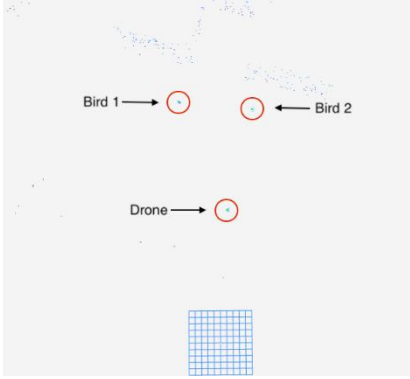
Results

Static Background Removal

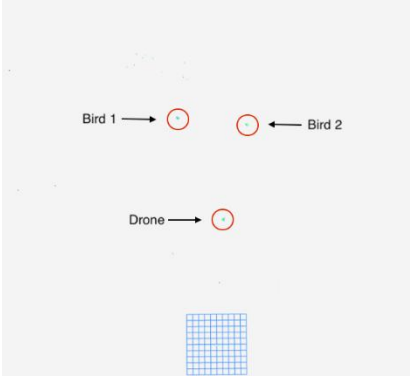
- 3D density threshold filtering requires a training phase
  - Lowest frame rate and highest computational cost
- Voxel confidence learning filtering adapts online
  - Moderate background suppression and throughput
- Occupancy ratio filtering provides
  - Strong background removal
  - Highest frame rate
  - Clean foreground with minimal residual clutter



3D Density Threshold



Voxel Confidence Learning



Occupancy ratio

➔ Occupancy ratio filtering for robust background suppression and real-time performance

Results

Object Detection

- Detection performance evaluation on simulated and real-world datasets
- Labelled dataset from manual annotation of captured frames in SUSTechPOINTS
- The simulated dataset with controlled forest environment with airborne agents
- The real-world dataset with outdoor LiDAR recordings under varying conditions
- Performance metrics
  - Precision
  - Recall
  - F1 score for multiple IoUs

Algorithm	Precision	Recall	F1 Score	TP	FP	FN
DBSCAN	**0.9079**	**0.9114**	**0.9097**	1173	119	114
HDBSCAN	0.6609	0.7786	0.7149	1002	514	285
K-Means	0.7664	0.8539	0.8078	1099	335	188
Mean Shift	0.7472	0.5672	0.6449	730	247	557

Table 1 - Simulated Dataset (IoU = 0.20)

Algorithm	Precision	Recall	F1 Score	TP	FP	FN
DBSCAN	**0.9097**	**0.9820**	**0.9445**	14879	1477	273
HDBSCAN	0.6105	0.5707	0.5899	8648	5518	6504
Mean Shift	0.4554	0.3044	0.3649	4613	5516	10539
K-Means	0.2998	0.2393	0.2661	3626	8470	11526

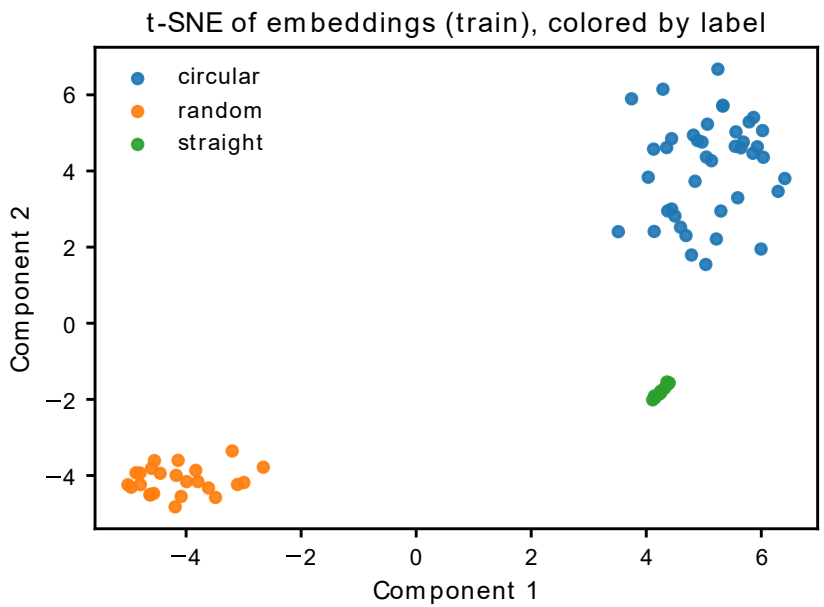
Table 2 - Real-World Dataset (IoU = 0.20)

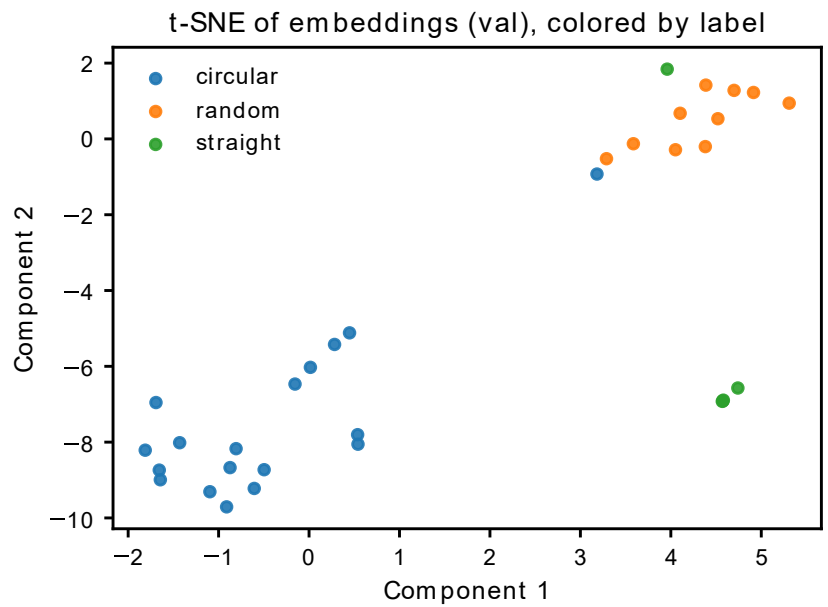
➔ DBSCAN with consistent high detection performance

Results - Object Classification

Latent Space Analysis for Supervised Reference Experiment

- Supervised contrastive learning for motion patterns separation
- Clear latent-space separation of trajectory types
- Preservation of coarse motion structure in validation embeddings





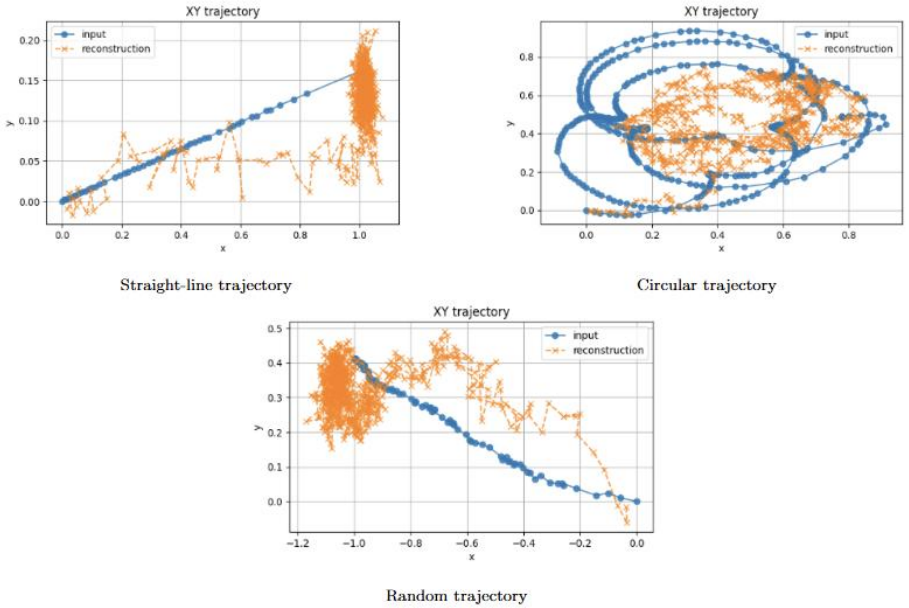
t-SNE Projections for train and val dataset

➔ Supervised latent separation with limited generalization to unseen trajectories

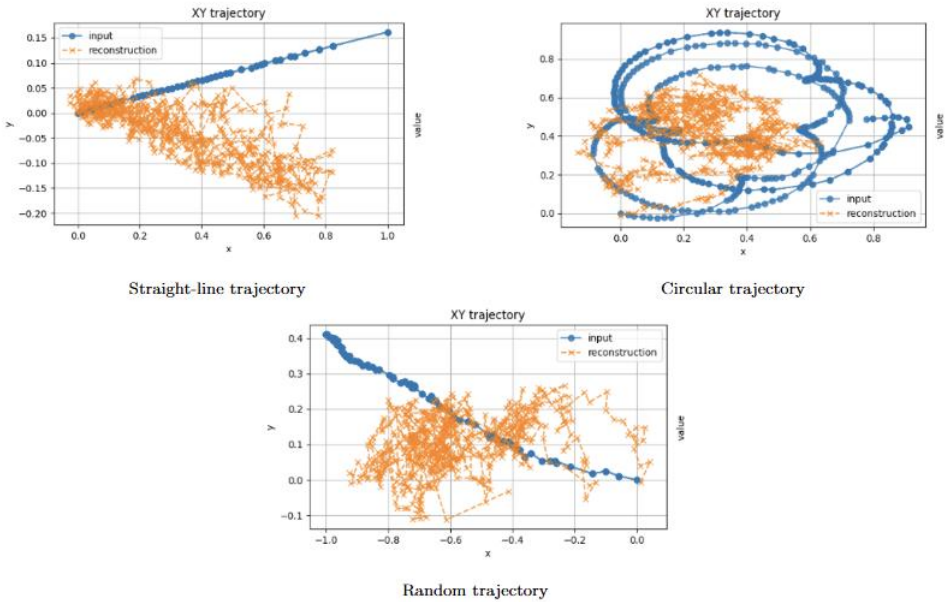
Results - Object Classification

Reconstruction Results for Unsupervised Centroid-Based Representation

- Model not able to reconstruct trajectories
- Directional motion trends partially encoded



Reconstruction for Repeat-Position Padding



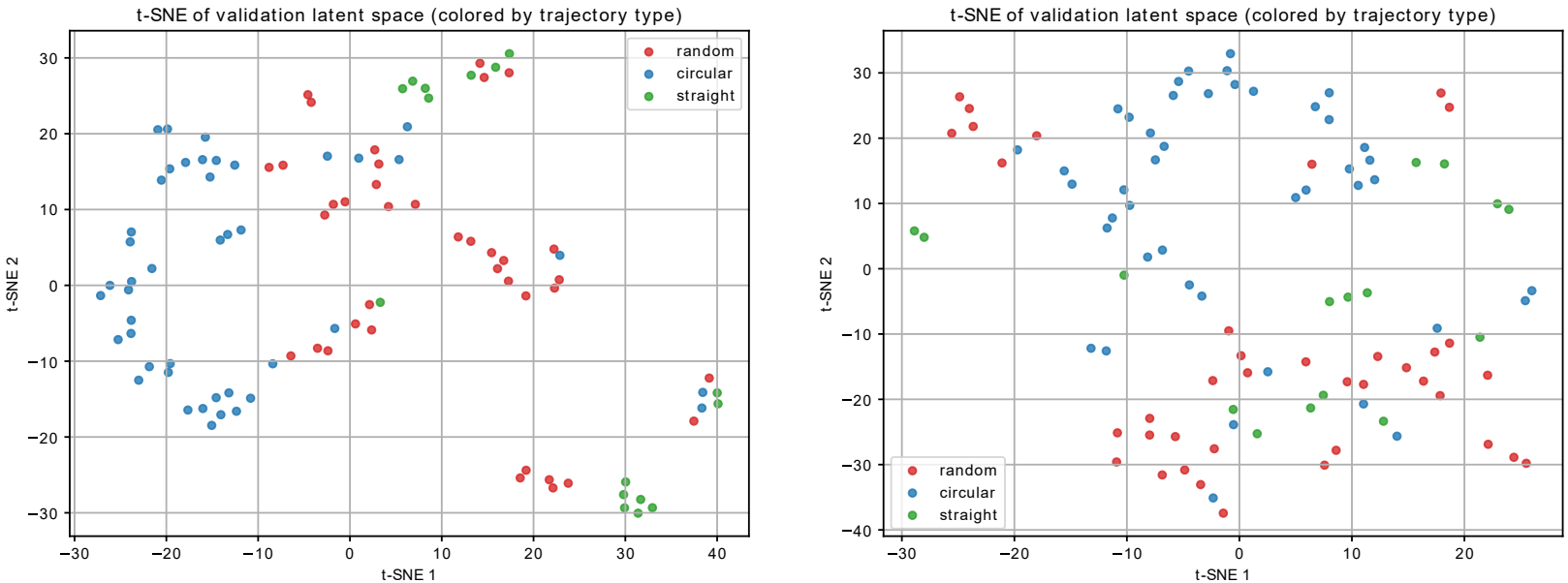
Reconstruction for Mirrored Padding

➔ Centroid-based autoencoder fails to reconstruct motion trajectories accurately

Results - Object Classification

Latent Space Analysis for Unsupervised Centroid-Based Representation

- Latent embedding analysis via t-SNE on synthetic trajectories
- Partial ordering of motion patterns
- Significant overlap among straight, circular, and random trajectories
- Weak and unstable separation across runs



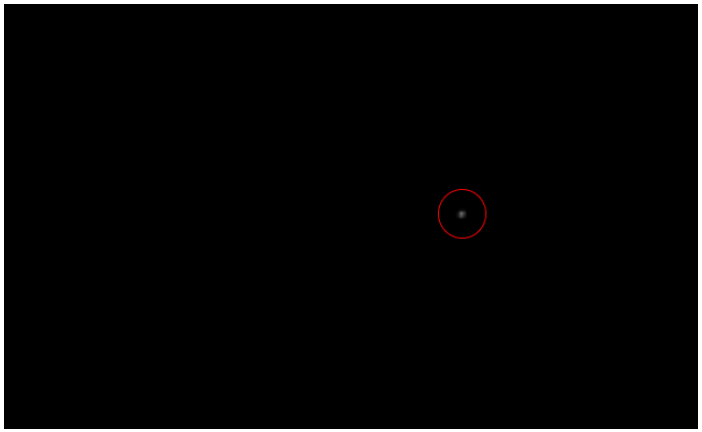
t-SNE Projections for Last-Position and Mirroring Padding

➔ Latent embeddings should not be interpreted as faithful motion representations

Results - Object Classification

Reconstruction Results for Model A

- Learning objective dominated by empty space reconstruction
- Model converges to trivial solutions under extreme sparsity
- Sparse occupied regions are not recovered
- Reconstruction loss does not reflect true occupancy quality



Gt-occupancy

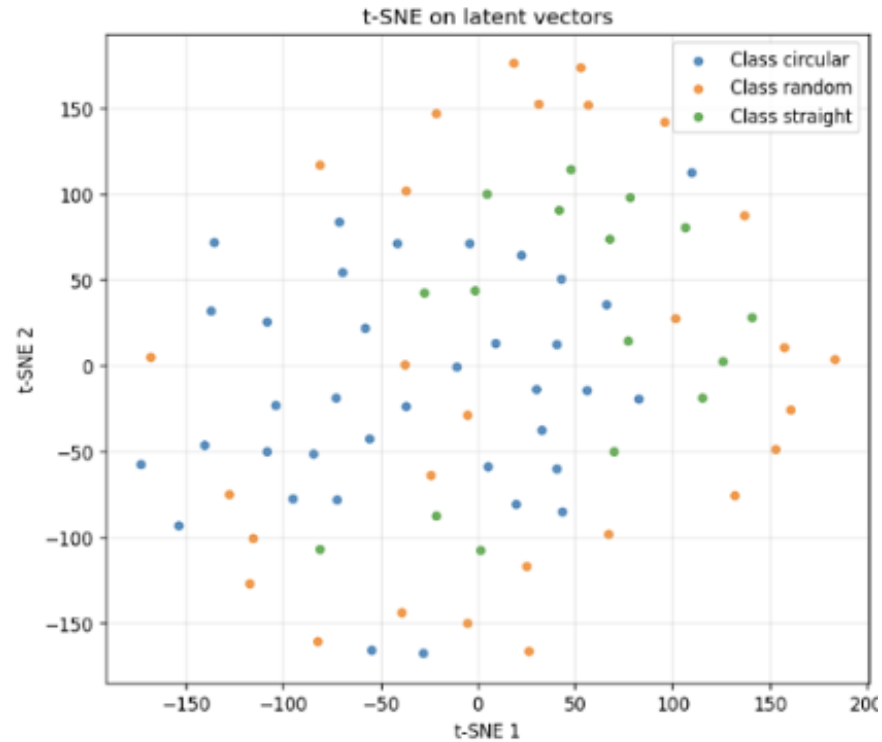
Predicted occupancy

➔ Reconstruction-based learning fails under sparse spatio-temporal occupancy

Results - Object Classification

Latent Space Analysis for Model A

- Spatio-temporal BEV sequences preserve full spatial structure
- Three-dimensional convolutional autoencoder trained for reconstruction
- Learning objective dominated by empty space reconstruction
- Model converges to trivial solutions under extreme sparsity



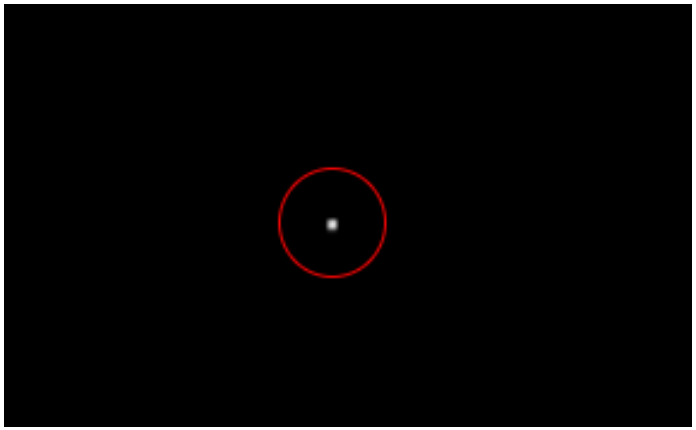
t-SNE Projections for Model A

➔ Learned representations do not encode discriminative motion behaviour

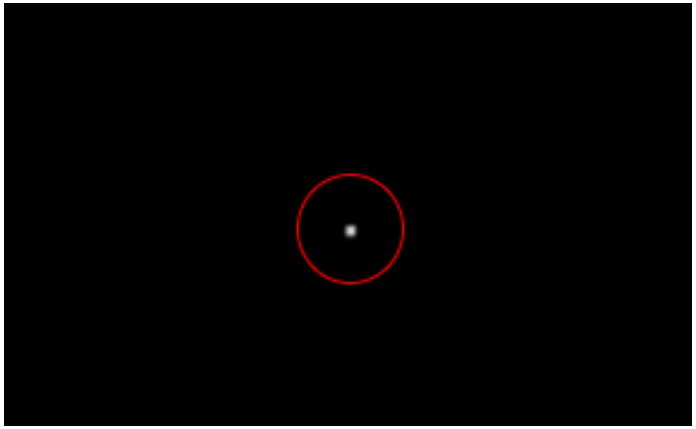
Results - Object Classification

Reconstruction Results for Model B

- Model predicts occupied regions instead of full reconstruction
- Sparse motion regions are recovered consistently
- Background regions remain suppressed
- No collapse to empty-space prediction
- Training remains stable under extreme class imbalance



Gt-occupancy



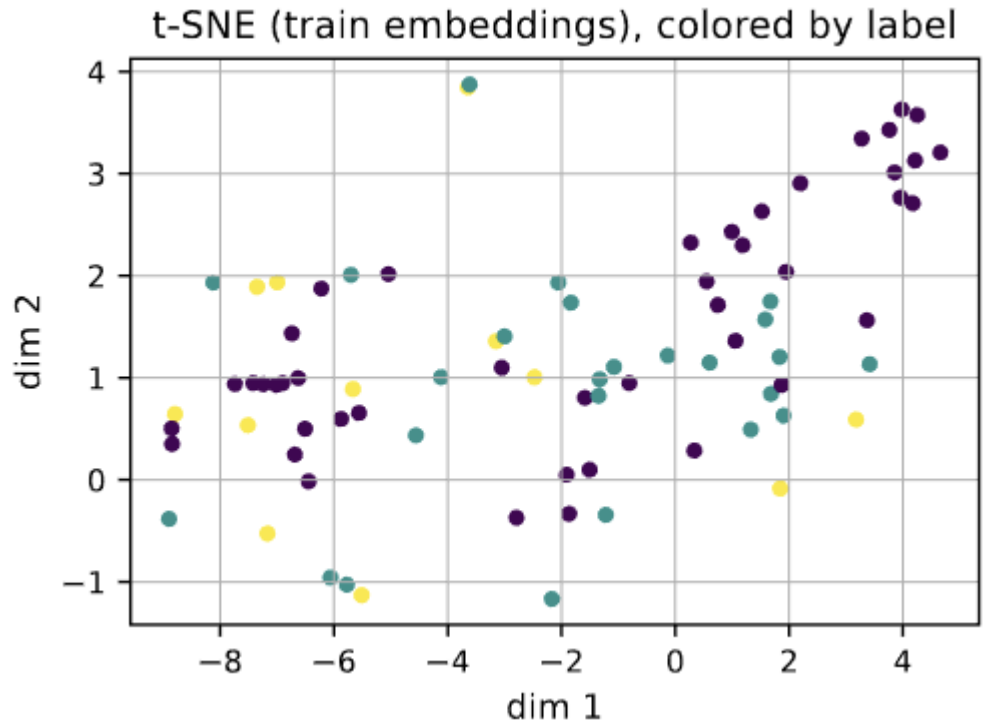
Predicted Occupancy

➔ Task-driven learning enables reliable recovery of sparse motion structure

Results - Object Classification

Latent Space Analysis for Model B

- Latent features extracted from intermediate network layers
- t-SNE shows locally structured feature groupings
- Similar motion patterns form neighbourhoods
- Significant overlap remains between trajectory types



t-SNE Projections for Model B

➔ Model B produces feature representations shaped by the occupancy detection objective

Discussion and Future Work

Summary

- Preprocessing determines downstream performance
- Adaptive filtering preserves small airborne targets
- DBSCAN provides stable detection
- Sparse data limits classification
- Reconstruction fails under sparsity

Conclusion

- Severe sparsity limits trajectory information
- Lack of labels restricts evaluation for classification
- Tracking fragmentation reduces scalability

Future Work

- Multi-LiDAR for longer trajectories
- Improved NIR imaging for reliable labelling
- Learning-based detection with supervision

Thank you!